

Fermi–Amaldi-Based Local Mixing Function

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(Dated: March 16, 2026)

I. MAIN TEXT

Consider the exact exchange energy density (closed-shell):

$$e_x^{\text{exact}}(\mathbf{r}) = -\frac{1}{4} \int \frac{|\gamma(\mathbf{r}; \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}', \quad (1)$$

where

$$\gamma(\mathbf{r}, \mathbf{r}') = 2 \sum_{i=1}^{N/2} \varphi_i^*(\mathbf{r}) \varphi_i(\mathbf{r}') \quad (2)$$

is one-electron reduced density matrix [1, 2], and the Fermi–Amaldi [3, 4] energy density expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N ,

$$e_x^{\text{FA}}(\mathbf{r}) = -\frac{1}{2N} \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (3)$$

For one-orbital systems (one-electron and two-electron singlet), two quantities are equal for all values of $\mathbf{r}$ [5–7]:

$$e_x^{\text{exact}}(\mathbf{r}) = e_x^{\text{FA}}(\mathbf{r}) \quad N \leq 2 \quad \text{for all } \mathbf{r}. \quad (4)$$

For systems with more than one molecular orbital, the spatial behavior of both quantities needs to be studied closer to establish a relation similar to Eq. (4). This is conveniently done by analyzing the corresponding exchange holes.

The exchange hole is defined via

$$e_x(\mathbf{r}) = \frac{1}{2} \int \frac{\rho(\mathbf{r})\rho_x(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \quad (5)$$

leading to expressions for the exact and the Fermi–Amaldi exchange holes, respectively:

$$\rho_x^{\text{exact}}(\mathbf{r}, \mathbf{r}') = -\frac{1}{2} \frac{|\gamma(\mathbf{r}, \mathbf{r}')|^2}{\rho(\mathbf{r})}, \quad (6)$$

$$\rho_x^{\text{FA}}(\mathbf{r}') = -\frac{1}{N} \rho(\mathbf{r}'). \quad (7)$$

When $N > 2$, we cannot establish an unambiguous inequality between these exchange holes for all $\mathbf{r}$. Although the Cauchy–Schwarz inequality states that

$$\rho(\mathbf{r})\rho(\mathbf{r}') \geq |\gamma(\mathbf{r}, \mathbf{r}')|^2 \quad N > 2 \quad \text{for all } \mathbf{r}, \quad (8)$$

the Fermi–Amaldi prefactor, $1/N$, decays faster than the constant prefactor, $1/2$, of the exact exchange hole. Therefore, there are some ranges of $|\mathbf{r} - \mathbf{r}'|$ dominated by one hole and others dominated by another.

As such, in the limit of infinitesimally small interelectronic distances ($r \rightarrow r'$) we have:

$$|\gamma(\mathbf{r}, \mathbf{r})|^2 = \rho(\mathbf{r})^2, \quad (9)$$

consequently, the depths of the exchange holes are

$$\rho_x^{\text{exact}}(\mathbf{r}, \mathbf{r}) = -\frac{\rho(\mathbf{r})}{2} \quad (10)$$

and

$$\rho_x^{\text{FA}}(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{N} \quad (11)$$

thus, for $N > 2$, the well of the exchange hole is dominated by the exact exchange hole.

The long-range behavior of the molecular orbital [8] is given by

$$\phi_i(\mathbf{r}) \sim \exp \left[-(-2\varepsilon_i)^{1/2} r \right], \quad (12)$$

where ε_i is the corresponding eigenvalue. Thus, in the limit of infinitely large interelectronic distances ($|\mathbf{r} - \mathbf{r}'| = \infty$), the electron density is given by:

$$\rho(\mathbf{r}) \sim \exp \left[-2(-2\varepsilon_i)^{1/2} r \right] \quad (13)$$

and the one-electron reduced density matrix is given by:

$$\gamma(\mathbf{r}, \mathbf{r}') \sim \exp \left[-(-2\varepsilon_i)^{1/2} r \right] \exp \left[-(-2\varepsilon_i)^{1/2} r' \right]. \quad (14)$$

Therefore, at large distances from the reference electron, both exchange holes decay euivalently.

II. SCRAP

A local hybrid exchange functional is given by

$$E_x^{\text{LH}} = \int \{g(\mathbf{r})e_x^{\text{exact}}(\mathbf{r}) + [1 - g(\mathbf{r})]e_x^{\text{SL}}(\mathbf{r})\} d\mathbf{r}, \quad (15)$$

where the integrand is a mixture of two exchange density energies weighed by a local mixing function (LMF), $g(\mathbf{r})$, which is position-dependent. The e_x^{SL} and e_x^{exact} quantities are a semi-local and the exact exchange energy densities, respectively.

A particular case of a one-electron limit, i.e., the asymptotic limit ($r \rightarrow \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r \rightarrow \infty} \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} = 1 \quad (16)$$

since both energy densities approach $-1/2r$ as $r \rightarrow \infty$.

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- [1] V. N. Staroverov, Density-functional approximations for exchange and correlation, in *A Matter of Density* (John Wiley & Sons, Ltd, 2012) Chap. 6, pp. 125–156.
 - [2] R. Parr and Y. Weitao, *Density-Functional Theory of Atoms and Molecules*, International Series of Monographs on Chemistry (Oxford University Press, 1989).
 - [3] E. Fermi and E. Amaldi, Le orbite ∞ s degli elementi, *Mem. Reale Accad. Italia* **6**, 117 (1934).
 - [4] P. Bénilan, J. A. Goldstein, and G. R. Rieder, The fermi-amaldi correction in spin polarized thomas-fermi theory, in *Differential Equations and Mathematical Physics*, Mathematics in Science and Engineering, Vol. 186 (Elsevier, 1992) pp. 25–37.
 - [5] I. P. Bosko and V. N. Staroverov, Derivation and reinterpretation of the fermi–amaldi functional, *J. Chem. Phys.* **159**, 131101 (2023).
 - [6] I. P. Bosko and V. N. Staroverov, Exchange energies and density functionals for systems of fermions of arbitrary spin, *Phys. Rev. A* **108**, 052803 (2023).
 - [7] P. W. Ayers, R. C. Morrison, and R. G. Parr, Fermi-amaldi model for exchange-correlation: atomic excitation energies from orbital energy differences, *Mol. Phys.* **103**, 2061 (2005).
 - [8] M. M. Morrell, R. G. Parr, and M. Levy, Calculation of ionization potentials from density matrices and natural functions, and the long-range behavior of natural orbitals and electron density, *The Journal of Chemical Physics* **62**, 549 (1975).